Economic Systems xxx (xxxx) xxxx

Contents lists available at ScienceDirect

Economic Systems

journal homepage: www.elsevier.com/locate/ecosys

Detecting contagion in Asian exchange rate markets using asymmetric DCC-GARCH and R-vine copulas

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ARTICLE INFO

JEL classification:
C32
C51
E42

Keywords:
Exchange rate contagion
Asian financial crisis
Copula functions
DCC-GARCH models

ABSTRACT

This study uses asymmetric DCC-GARCH models and copula functions to study exchange rate contagion in a group of twelve Asia-Pacific countries. Using daily data between November 1991 and March 2017, we show that extreme market movements are mainly associated with the high degree of interdependence registered by countries in this region. Evidence of contagion is scarce. Asymmetries do not appear to be important. Specifically, currency co-movements are statistically identical during times of extreme market appreciation and depreciation, indicating that phenomena such as the fear of "appreciation" do not appear to be relevant in the region's foreign exchange markets.

1. Introduction

This paper studies exchange rate contagion in Asia-Pacific markets between 1991 and 2017 following an R-vine copula approach.[1] As in Forbes and Rigobon (2001), contagion consists of a significant increase in cross-market linkages after the occurrence of a shock in a country's financial market. Defining contagion this way offers two important advantages. First, it allows distinguishing between temporal and permanent mechanisms for the transmission of crises, facilitating the implementation of macro-prudential policies. Second, it provides a useful frame work for empirically testing contagion in a very general setup. Here, we employ the tail dependence criterion proposed by Cherubini et al. (2004) and used in related studies including Czado et al. (2012), Loaiza-Maya et al. (2015a, b) and Cubillos-Rocha et al. (2019).

Contagion during currency crises has been extensively studied. Much of the ample empirical literature emphasizes its geographical component (see Lee and Kim, 1993; Forbes and Rigobon, 2001; Dungey et al., 2006; Lucey and Voronkova, 2008; Arouri et al., 2012). Currency crises tend to be regional, as they affect countries in geographical proximity (Glick and Rose, 1999; De Gregorio and Valdes, 2001; Beirne and Gieck, 2014; Grabowski and Welfe, 2016; Lee et al., 2016).

Studying developed and Asia-Pacific exchange rate markets from the 1990s onwards is interesting for various reasons. First, the

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[1] We use R-vines in our study for two main reasons. The first is that most related studies using copula functions follow a vine approach. Hence, in order to make our results more comparable to those of the closest papers in the literature, we maintain their econometric approach. Second, the most general vines are regular vines. C-vines and D-vines are particular cases of R-vines. A few recent papers in this strand of the literature have used dynamic copulas (with and without regime switching, nonparametric approaches and wavelet coherence analysis. An interesting extension of our study would be to use these other approaches in the context of exchange rate co-movements in Asia.

https://doi.org/10.1016/j.ecosys.2019.100717
Received 22 August 2018; Received in revised form 13 November 2018; Accepted 21 December 2018
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Please cite this article as: Jose E. Gomez-Gonzalez and Wilmer Rojas-Espinosa, Economic Systems, https://doi.org/10.1016/j.ecosys.2019.100717

term “contagion” in financial markets began to be used after the Asian banking and currency crises of 1997 (Claessens and Forbes, 2001). There is relative consensus in the literature that these are benchmark events for studies on interdependence and spillovers among financial markets. Hence, providing further evidence for a better understanding of contagion during those episodes is always useful. Second, the Asia-Pacific region’s importance in the global economy is growing. Estimates of the IMF indicate that, by 2030, Asia’s economy will be larger than those of the United States and the European Union combined. The countries in this region are increasingly gaining importance in international financial markets and, therefore, their potential of transmitting volatility shocks to other emerging and developed economies is also growing (Engle et al., 2012). Finally, currency markets in the region have undergone important reforms and transformations over the last two decades. Thus, the data presents sufficient heterogeneity to identify interdependence and contagion appropriately.

The results show that the Malaysian ringgit is central in the region’s exchange rate network. On the contrary, the Japanese yen and the New Zealand dollar are the least integrated currencies in the network. In contrast to recent studies that have found evidence of contagion in Latin American and East European countries, this paper shows little evidence of contagion for the Asia-Pacific region. Exchange rate co-movements are mainly due to high interdependence between countries, as in Forbes and Rigobon (2001).

We find that exchange rate dependencies in Asia behave quite similarly during periods of large appreciation and depreciation. In other words, our evidence suggests that the asymmetries in exchange rate contagion that have been encountered for other regions and other samples of countries (see, for instance, Levy-Yeyati et al., 2013) do not hold for our sample of Asian economies. This result may be due to the fact that central bank intervention is more symmetric in Asian countries than, for instance, in Latin American countries, where a “fear of appreciation” effect has been identified in the literature.

Using a large number of Asian economies and a long sample period covering the global financial crisis, we contribute to the literature on exchange rate contagion by showing that exchange rate co-movements in this region are mainly due to the fact that their foreign exchange rate markets are becoming more integrated. This result contrasts importantly with those of other similar papers that have shown evidence of contagion in other regions of the world such as Latin America (Loaiza-Maya et al., 2015a, b). Our results are in line with those of Forbes and Rigobon (2001), who show that during the Asian financial crisis of 1997, exchange rate interdependence increased for countries in the region without evidence of contagion. Additionally, our asymmetric DCC GARCH approach allows us to test for possible asymmetric co-movements during large appreciations and depreciations. The results indicate that co-movements are symmetric in this sense. This result, which is also in contrast with those of studies for other regions, shows that contagion exhibits an important regional component.

The fact that contagion has an important regional component provides important information for global investors. Specifically, it indicates that global hedging strategies require regional diversification. This finding complements those of Dimitriou and Kenourgios (2013) and Dimitriou et al. (2017), who show that correlation dynamics between currencies vary largely over time, increasing vulnerabilities during times of financial distress. Hence, there are higher portfolio diversification benefits during times of financial distress, since holding a diversified currency portfolio reduces systemic risk more during those times.

This paper is organized as follows. Section 2 presents the data and the descriptive statistics. Section 3 introduces the ARIMA-GARCH best models and the DCC specification to account for asymmetric effects to obtain the standardized residuals. The Sklar’s theorem is presented in Section 4 along with the R-vine and tail dependence definitions, which are used for empirical estimations in Section 5. The last section concludes.

2. Data and descriptive statistics

This document covers a synchronized period from 7 November 1991 to 16 March 2017 for daily closing values obtained from Bloomberg L.P. Exchange rate data is gathered for the twelve Asia-Pacific economies: Australia (AUD), Hong Kong (HKD), India (INR), Indonesia (IDR), Japan (JPY), Malaysia (MYR), New Zealand (NZD), Singapore (SGD), South Korea (KRW), Thailand (THB), Taiwan (TWD) and the Philippines (PHP). The nominal exchange rates are depicted in Fig. A1 in the Appendix A.

Some recent papers argue that results from hypothesis tests in finance are frequency dependent (Narayan and Sharma, 2015; Narayan et al., 2015). Therefore, the selection of the frequency of the data is not trivial in empirical studies in finance. Daily data has been shown to contain richer information than data at lower frequencies (see Kenourgios et al., 2016; Bannigidadmath and Narayan, 2016).

Table 1 presents the descriptive statistics for exchange rate returns.[2] Note that the means are positive and negative indicating devaluations and appreciations over the period of study. Return distributions are skewed, presenting higher mass on the right tails as well as the left tails. This fact illustrates that the currencies are affected by depreciations and appreciations. However, it appears to be the case that appreciations are more frequent within individual countries’ exchange rates and justify the exploration of asymmetries in interdependence and contagion.

Interestingly, excess Kurtosis is positive in all cases. The distribution exhibits platykurtic distributions, having fewer extreme values. This empirical fact indicates that most countries in the Asia-Pacific region exhibit few episodes of high exchange rate volatility, possibly associated with active central bank exchange rate intervention policies. However, considerable heterogeneity is observed between countries.

[2] We report information on returns rather than on exchange rates, as the former are covariance stationary, which the latter are not. The returns are computed as the logarithmic difference of nominal exchange rates. $r_{i,t} = ln(Y_{i,t}) - ln(Y_{i,t-1})$, (1) All exchange rates are computed as the number of units of each country’s currency that buy one US dollar.

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Table 1
Exchange rate returns; Descriptive statistics.

	AUD	HKD	INR	IDR	JPY	MYR	NZD	SGD	KRW	THB	TWD	PHP
Mean	−3.15e-06	1.18e-08	−0.0001	−0.0002	2.07e-05	7.32e-05	3.34e-05	2.70e-05	−6.08e-05	−4.76e-05	−2.35e-05	−9.81e-05
Median	0.0002	0.00	0.00	0.00	0.00	0.00	0.0001	0.00	0.00	0.00	0.00	0.00
Maximum	0.0827	0.0062	0.0347	0.2309	0.0705	0.0985	0.0451	0.0371	0.2034	0.0855	0.0323	0.0644
Minimum	−0.0729	−0.0028	−0.0610	−0.1999	−0.0547	−0.0770	−0.0673	−0.0302	−0.1525	−0.1779	−0.0333	−0.1085
Std.Dev.	0.0075	0.0003	0.0041	0.0135	0.0068	0.0050	0.0075	0.0035	0.0086	0.0059	0.0029	0.0061
Skewness	−0.2555	0.9591	−1.2494	−0.7468	0.3594	0.4196	−0.2234	0.1825	−0.2602	−3.9070	−0.3543	−1.1868
Kurtosis	12.543	29.377	24.502	84.303	8.5256	76.403	6.5774	12.061	104.41	154.53	18.458	39.589
Jarque-Bera	25086.5	192121.2	128704.5	1816225.	8528.2	1480114.	3570.1	22588.7	2824798	6323502.	65777.1	369271.9
Probability	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Kolmogorov-Smirnov	0.06	0.22	0.14	0.23	0.06	0.19	0.05	0.07	0.17	0.17	0.11	0.14
Probability	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 2
Best fit ARIMA models.

Country	ARIMA	AIC	BIC
Australia	(1,1,1)	−49368.54	−49348.16
Hong Kong	(1,1,1)	−113245.4	−113225.1
India	(0,1,2) with drift	−102750.9	−102723.7
Indonesia	(1,2,1)	−64371.53	−64351.15
Japan	(0,1,0)	−108383.1	−108376.2
Malaysia	(4,1,4)	−67904.48	−67843.34
New Zealand	(0,1,0)	−51445.11	−51438.31
Singapore	(0,1,3)	−60926.94	−60899.77
South Korea	(0,1,0)	−77876.19	−77869.4
Thailand	(1,1,1)	−95429.72	−95409.34
Taiwan	(0,1,0)	−103372.6	−103365.8
The Philippines	(1,1,1) with drift	−94573.04	94545.87

The Jarque-Bera test results provide evidence for rejecting the normal distribution hypothesis in all countries at conventional confidence levels. Additionally, the D-values for the Kolmogorov-Smirnov statistics are positive, providing further evidence that the returns are not normally distributed. Together, this suggests the importance of exploring alternative distributions that can better represent exchange rate returns and their dependence structures.

3. ARIMA-GARCH

The best fit ARIMA models are calculated according to BIC and AIC for each currency, as shown in Table 2. The standard GARCH specification is applied:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{2}$$

using a Student-t conditional distribution as proposed by Hansen (1994):

$$g(z|\eta,\ \lambda) = \begin{cases} bc\left(1 + \frac{1}{\eta-2}\left(\frac{bz+a}{1-\lambda}\right)^2\right)^{-(\eta+1)/2} & z < -a/b, \\ bc\left(1 + \frac{1}{\eta-2}\left(\frac{bz+a}{1-\lambda}\right)^2\right)^{-(\eta+1)/2} & z \geq -a/b, \end{cases} \tag{3}$$

where $2 < \eta < \infty$, and $-1 < \lambda < 1$. The constants a, b, and c are given by $a = 4\lambda c\left(\frac{\eta-2}{\eta-1}\right)$, $b^2 = 1 + 3\lambda^2 - a^2$ and $c = \frac{\Gamma\left(\frac{\eta+1}{2}\right)}{\sqrt{\pi(\eta-2)}\,\Gamma\left(\frac{\eta}{2}\right)}$.

3.1. DCC

Tse and Tsui (2000) and Engle (2002) introduced the dynamic conditional correlations (DCC) as an extension to the CCC model of Bollerslev (1990) to correct for constant correlation over time. The variance-covariance matrix is defined as:

$$H_t = D_t R_t D_t, \tag{4}$$

where $D_t = diag\ \sqrt{h_{i,j}}$. D_t is a diagonal matrix containing the conditional standard deviations on the leading diagonal; R_t is the conditional correlation matrix, its specification is formulated by Engle (2002) as $R_t = diag\ \{Q_t\}^{-1} Q_t\ diag\ \{Q_t\}^{-1}$. Q comes from a general MGARCH model where more complex positive definite multivariate GARCH models could be used for the correlation parametrizations.

$$Q = S \circ (u' - A - B) + A \circ \varepsilon_{t-1}\varepsilon'_{t-1} + B \circ Q_{t-1} \tag{5}$$

where $\circ$ is the Hadamard product; $A,\ B\ and\ (u' - A - B)$ are positive semidefinite and Q will be positive semidefinite.

Chiang et al. (2007) implemented a symmetric DCC-GARCH for eight daily Asian stock-return data series from 1990-2003. The DCC-GARCH specification in this document accounts for asymmetric effects as recommended by Cappiello et al. (2006), i.e. volatility increases more after a negative shock than after a positive shock of the same magnitude. This model is appropriate for measuring time-varying conditional correlations and the responses to news and innovations. The standard residuals for our model for each currency are displayed in Fig. A2 in the Appendix A.

The pseudo-sample associated with these residuals to uniform margins ($u_1,\ u_2 \in [0,1]$) can be used to construct currency pairs, as presented in Fig. A3 in the Appendix A. In this matrix, the lower half below the diagonal presents the level plots or contours and the upper half indicates the Kendall's Taus measure of all possible pairs. These uniform margins are used to analyze their joint distribution and ultimately their dependence structure.

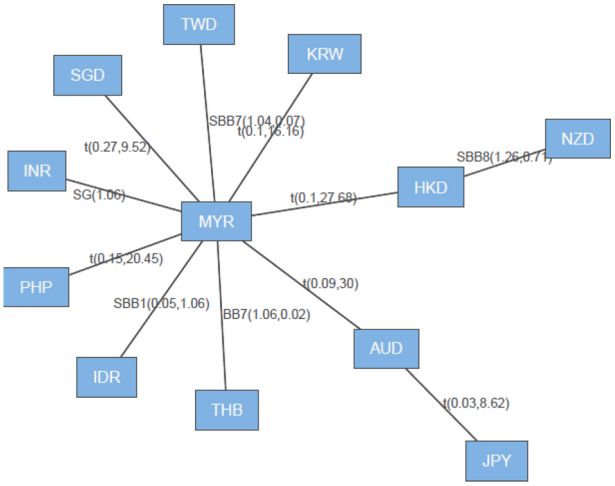


Fig. 1. Tree 1 in R-vine.

4. Sklar's theorem, 1959

Copula functions are useful for modelling multivariate dependence, especially when normality fails to be a good assumption for distributions, as in our case. Moreover, copulas facilitate isolating dependence between random variables from their marginal distributions.

Since the probability integral transform is invertible, the copula also describes the dependence between the original variables. Notably, in economics there is often more information about the marginal distributions of related variables than their joint distribution. Inasmuch as copulas can capture the dependence structure regardless of the form of the margins, a copula approach is likely very useful in econometrics.

Succinctly, let an n-dimensional distribution function H be decomposed into two parts, the marginal distribution functions F_i and the copula C. Let H be a joint distribution function with margins F_i. Then there exists a copula C such that for all x_i in $\bar{R}^n$ following the seminal work of Sklar (1959),

$$H(x_1. ..x_n) = C(F_1(x_1), ...,F_n(x_n)). \tag{6}$$

Let $F_1, \ldots, F_n$ be the distribution functions. If $F_1, \ldots, F_n$ are all continuous, then C is unique; otherwise, C is uniquely determined on $RanF_1 x \ldots RanF_n$. Conversely, if C is an n-copula and $F_1, \ldots, F_n$ are distribution functions, then the function H defined above is an n distribution function with margins $F_1, \ldots, F_n$ (see Joe, 1997; Nelsen, 2007).

4.1. R-vine

A flexible graphical method proposed by various authors (Joe, 1996; Bedford and Cooke, 2001, 2002; Kurowicka and Cooke, 2006) describes multivariate copulas as a dependency model for the distribution of certain pairs of variables conditional on a specified set of variables. Using this cascade of bivariate copulas, so-called pair copulas construction (PCC), the recognition of the needed pairs of variables and their set of conditional variables is facilitated by an array of trees.

Vines arrange the n(n − 1)/2 pair copulas of an n-dimensional PCC in n − 1 linked trees. In general, the structure captures conditional dependencies in higher trees and the order of the nodes is selected in such a way that the strongest pairwise dependencies are captured in the first tree.

The specification of the R-vine copula has the following arguments: $(\mathbf{F}, \nu, B)$ is an R-vine copula specification if $\mathbf{F} = (F_1, ..., F_n)$ is a vector of continuous invertible distribution functions; as Sklar's theorem illustrated in Eq. (6), ν is an n-dimensional R-vine and $B = \{B_e | i = 1, ..., n-1;\ e \in E_i\}$ is a set of copulas, with B_e being a bivariate copula $(X_{C_{e,a}}, X_{C_{e,b}})$.

The R-vine decomposition of a multivariate density is:

$$f(x_1, ...,x_n) = \prod_{k=1}^{n} f_k(x_k) \times \prod_{i=1}^{n-1} \prod_{e \in E_i} c_{C_{e,a},\, C_{e,b}\,|D_e}(F_{C_{e,a}|D_e}(x_{C_{e,a}}|x_{1De}, \ldots,x_{nDe}),\ F_{C_{e,b}|D_e}(x_{C_{e,b}}|x_{1De}, \ldots,x_{nDe})), \tag{7}$$

where f_k are the marginal densities $k = 1, \ldots, n$, $c_{C_{e,a},\, C_{e,b}\,|D_e}$ is the copula density of B_e for edge $e = \{a, b\}$ and $x_{i_{D,e}} \in D_e$ for $i = 1, \ldots, n$. Joe (1996) describes the strategy to obtain the conditional distribution $F_{C_{e,a}|D_e}(x_{C_{e,a}}|\bullet)$ and $F_{C_{e,b}|D_e}(x_{C_{e,b}}|\bullet)$.

4.2. Tail dependence

The concept of tail dependence is used as an advanced device to identify contagion among currencies when extreme values occur.

This measure is only copula-based and can thus be used in the parametrization of copulas. Joe (1997) defines this as the amount of dependence in the upper-quadrant tail or lower-quadrant tail of a bivariate distribution.

By definition:

$$\lambda_L = lim_{u \searrow 0} \frac{C(u, u)}{u} \tag{8}$$

and

$$\lambda_U = lim_{u \nearrow 1} \frac{1 - 2u + C(u, u)}{1 - u} \tag{9}$$

where λ_L is the lower tail and λ_U is the upper tail dependence for a given bivariate copula family and parameter(s).

A copula is said to have a lower (upper) tail dependence if $\lambda_L \neq 0$ ($\lambda_U \neq 0$). The lower tail dependence measure (λ_L) is the limiting value of $\frac{C(u,u)}{u}$, which is the conditional probability $Pr[U_1 < u | U_2 < u] (= Pr[U_2 < u | U_1 < u])$, and the upper tail dependence (λ_U) is the conditional probability $Pr[U_1 > u | U_2 > u] (= Pr[U_2 > u | U_1 > u])$.

5. Estimation

This document uses the automated strategy of jointly searching the appropriate R-vine tree structure, the pair copula families and their parameter values, formulated by Dissmann et al. (2013). Succinctly, the method consists of selecting exchange rate pairs, finding the best bivariate copula families for each pair using information criteria, and estimating the corresponding parameters by Maximum Likelihood.

Table A1 in the Appendix A presents the estimation results. The first column displays the tree number. The edge, shown in the second column, presents pair currencies (unconditional for the first tree and conditional for the rest of them). The best types of copula fit are presented in the third and fourth columns. The other three columns show information on the value of the parameters and Kendall's Tau. Note that a t distribution is frequently the most adequate for the diff ;erent exchange rate pairs. Other distributions that frequently appear as adequate belong to the SBB family.

The Malaysian ringgit maximizes the sum of the absolute empirical Kendall's Tau (see Fig. A3 in the Appendix A).

In other words, the first spanning tree (Fig. 1) presents Malaysia's ringgit as the main node, while the rest of the currencies are its immediate edges (except for Japan's yen and New Zealand's dollar). These two currencies are connected to the network through other currencies. In the case of the Yen, its connection is mediated by the Australian Dollar. Meanwhile, New Zealand's dollar is connected through Hong Kong's currency.

These are interesting results, which are useful for investors, as they highlight important issues regarding diversification opportunities. Specifically, we show that Malaysia is the main shock transmitter. However, volatility transmission first flows to most countries of the Asia-Pacific region, including Australia and Hong Kong, while Japan and New Zealand are only indirectly affected by their connection to other countries in the network.

Since we want to identify contagion among currencies when extreme values occur, the tail values are calculated using definitions 8 and 9. The upper tail dependence of the twelve Asian exchange rates is displayed in the top right panel of Table 3. Upper tail dependence is associated with currencies' co-movement for large depreciations. On the other hand, lower tail dependence indicates large appreciations, as shown in the bottom panel of the same table.

Table 3 shows our main results regarding tail dependence. Note that the tail values are mainly symmetric and their probabilities are lower than 1% (however, different from zero in most cases). The fact that values are symmetric for most pairs implies that interdependence is similar in times of extreme currency appreciations and depreciations with respect to the US dollar. This result, which is in contrast with those of the majority of studies (see Loaiza-Maya et al., 2015a, b), shows that phenomena such as the "fear of appreciation" (see Levy-Yeyati et al., 2013) are not present in the Asia-Pacific region. In other words, while central banks in these

Table 3
Tail dependence

	AUDHKDINR	IDR	JPYMYRNZDSGDKRWTHB	TWD	PHP
AUD	2.70e-10 0.00	0.00	2.15e-3 0.005.14e-6 1.01e-2 4.77e-6 0.00	2.69e-4	1.35e-2
HKD	0.00 0.003.67e-6		9.32e-5 0.000.000.001.73e-2 6.63e-4	1.56e-3	7.29e-2
INR	0.000.00 9.18e-6		0.001.27e-4 0.000.006.39e-6 3.75e-5	0.00	6.40e-4
IDR	0.003.67e-6 9.18e-6		0.003.54e-6 3.65e-6 4.93e-5 0.001.40e-3	0.00	0.00
JPY	2.15e-3 9.32e-5 4.19e-2 0.00		0.00 8.08e-6 0.001.97e-3 0.00	0.00	1.77e-5
MYR	0.000.00 1.27e-4 3.54e-6		0.00 0.000.000.002.57e-5	6.91e-8	0.00
NZD	5.14e-6 0.00 0.003.65e-6		8.08e-6 0.00 1.12e-2 1.35e-5 0.00	3.88e-2	4.25e-5
SGD	1.01e-2 1.16e-2 0.004.93e-5		0.000.00 0.005.60e-6 7.09e-4	1.16e-4	6.07e-7
KRW	4.77e-6 0.00 6.39e-6 2.91e-14 1.97e-3 0.00		1.35e-5 5.60e-60.0075	0.0003	0.0015
THB	0.006.63e-4 3.75e-5 1.40e-30.002.57e-5 1.88e-2 7.09e-4 0.0075			0.0011	0.0321[a]
TWD	2.69e-4 1.56e-3 0.000.000.000.005.34e-7 1.16e-4 0.00030.0011				5.92e-5
PHP	1.35e-2 3.33e-14 6.40e-4 0.001.77e-5 7.77e-2 4.25e-5 7.52e-2 0.00150.0321[a]			0.0507[b]	

[a] if λ_L or $\lambda_U \geq$ Pr[1%].
[b] if λ_L or $\lambda_U \geq$ Pr[5%].

Table 4
Multivariate specification tests for the standardized residuals.

	Null Hypothesis	Lags	Statistics	P-Value
Portmanteau	No autocorrelation	300	24795.691	0.215
LM (square residuals)	No MGARCH effect	100	94545.87	0.242

countries conduct exchange rate intervention, it appears that these interventions do not depend on whether the exchange rate is devaluating or revaluating. In this sense, the "fear of appreciation" seems to be a common phenomenon mostly in Latin American and East European economies.

Pair currencies with both tails equal to zero ($\lambda_L = 0 and \lambda_U = 0$) as in Gaussian copulas are: Australian dollar-(Indian rupee, Indonesian rupiah, Malaysian ringgit, Thailand baht), Hong Kong dollar-(Indian rupee, Malaysian ringgit, New Zealand dollar), Indian rupee-(New Zealand dollar, Singapore dollar, Taiwan new dollar), Indonesia rupiah-(Japanese yen, Taiwan new dollar, Philippine peso), Japan yen-(Malaysian ringgit, Singapore dollar, Thailand baht, Taiwan new dollar) and Malaysian ringgit-(New Zealand dollar, Singapore dollar, South Korean won). In other words, for these cases there is no contagion from large appreciations or depreciations.

This finding then encourages the same results as Forbes and Rigobon (2001), Basu (2002) and Bordo and Murshid (2000). In other words, transmission mechanisms are fairly stable among these currencies and little contagion can be spread. One interpretation is that the Asian financial crisis of 1997-1998 made the implementation of various types of policy coordination necessary, the main purpose of which was to internalize the externalities and spill-over effects that arise from economic regionalism. Moreover, there was growing resentment towards the International Monetary Fund (IMF) and U.S. handling of the crisis, which intensified the interest in the East Asian Economic Group (EAEG), taking the form of the ASEAN Plus Three (APT) framework.

Central banks and governments in East Asia agreed to create a regional self-help mechanism for the effective prevention and management of financial crises, which included regional economic surveillance led by the Economic Review and Policy Dialogue (ERPD), a regional liquidity support facility called the Chiang Mai Initiative (CMI), and local-currency bond market development. Kawai (2005) suggests that the lesson from the Asian financial crisis was a clear need for effective prevention, management and resolution of financial crises and contagion, a thought that seems to justify the tail dependence values in the upper and lower cases equal to zero.

The Philippine peso-Taiwan new dollar only has a lower tail or appreciation probability greater than 5%. The ASEAN currencies Philippine peso-Thailand baht have symmetric upper and lower tails greater than 1%.

The results for the tests on standardized residuals indicate that our empirical specifications are adequate (see Table 4).

We perform two types of robustness tests. One deals with country exclusion and the other with the reduction of the sample period. Regarding the first, we excluded constructed R-vines, excluding the countries with less developed foreign exchange markets in our sample, and observed that the results were qualitatively identical. Particularly, no evidence of contagion was encountered. Regarding the second, the sample period was shortened to exclude the recent international financial crisis. Data were considered only for the period between 2009 and 2018. The results were qualitatively identical as well.

6. Conclusions

This paper studied exchange rate contagion in the Asia-Pacific region. Using daily data spanning the period November 1991 to March 2017, together with DCC-GARCH models and copula functions, the paper explored the behavior of interdependence in times of extreme market appreciations and depreciations and their potential asymmetric effects.

In this context, contagion is considered as a situation in which exchange rate cross-market linkages significantly increase after the occurrence of a shock in a country's financial market. The sample contains countries from a unique region, as evidence shows that currency crises tend to be regional, i.e., they affect countries in geographical proximity.

The R-vine copula approach followed in this study allows the identification of the best bivariate copula family for each exchange rate pair, as it permits the estimation of tail dependence coefficients for extreme exchange rate appreciations and depreciations. The results show that, in most cases, a t-copula or a copula pertaining to the SBB family is the best fit. The network centrality of the Malaysian ringgit is identified, while the Japanese yen and the New Zealand dollar are the most isolated currencies in the sample. These results provide important information for investors interested in portfolio balancing with assets from countries of the Asia-Pacific region.

In contrast with other studies on contagion that follow a similar approach, the results of this study indicate that important asymmetries are not encountered and evidence of contagion is scarce. This result, which is in line with those of Forbes and Rigobon (2001), suggests that exchange rate co-movements in the region's currencies are due to high interdependence between countries.

Our results suggest that co-movements in Asian economies are mainly derived from their higher connectedness and integration over time. Foreign exchange markets in this region have developed strongly, as currencies of Asian countries have become increasingly important in the composition of global investment portfolios and in international trading. When comparing our results to those of related studies using countries from other regions, the importance of regional components in contagion becomes evident. More specifically, exchange rate contagion is more likely to occur within countries of regions with less developed financial markets, like for example Latin America. This result highlights the importance of regional diversification in constructing global hedging strategies in foreign exchange markets.

Appendix A

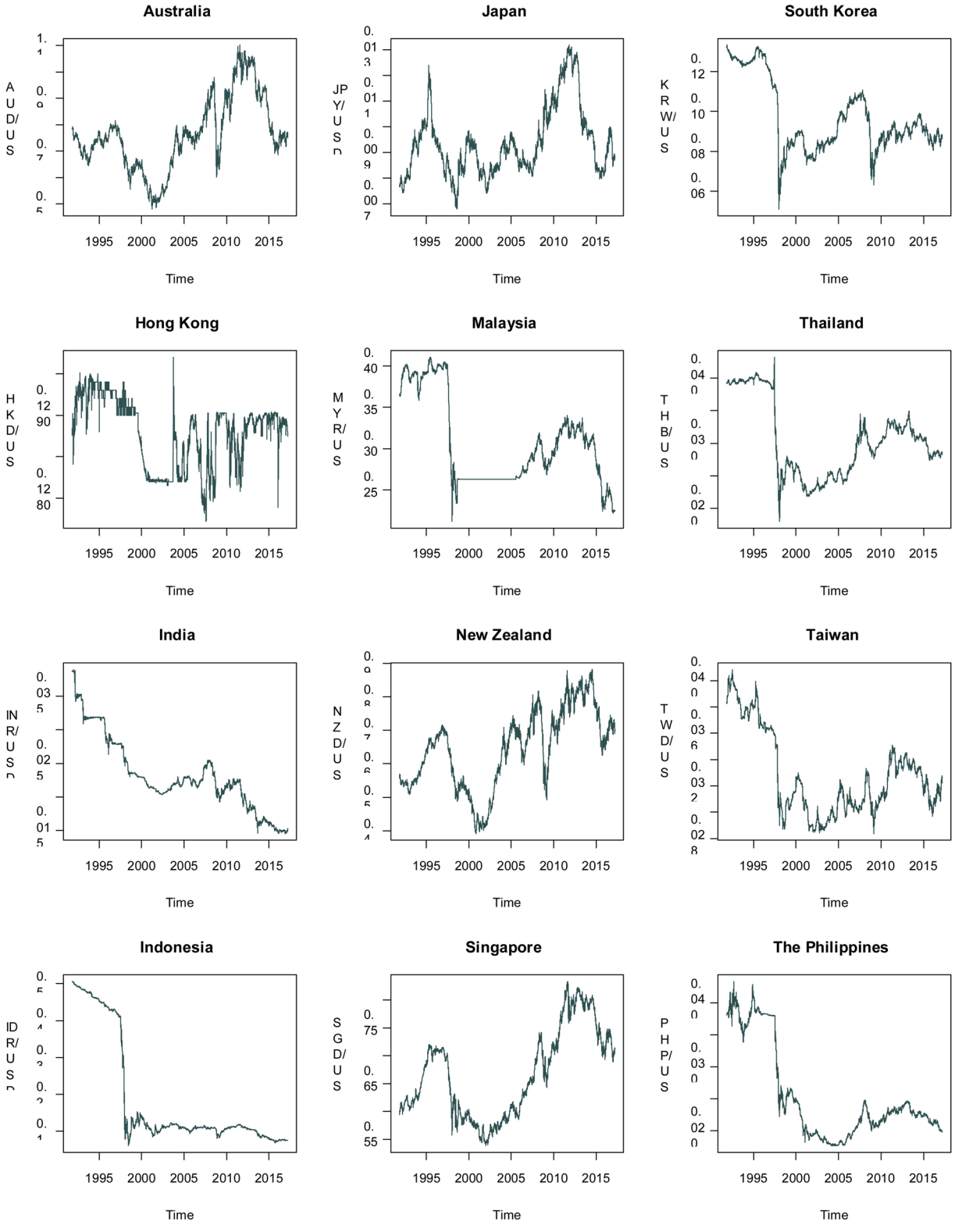


Fig. A1. Nominal exchange rates.

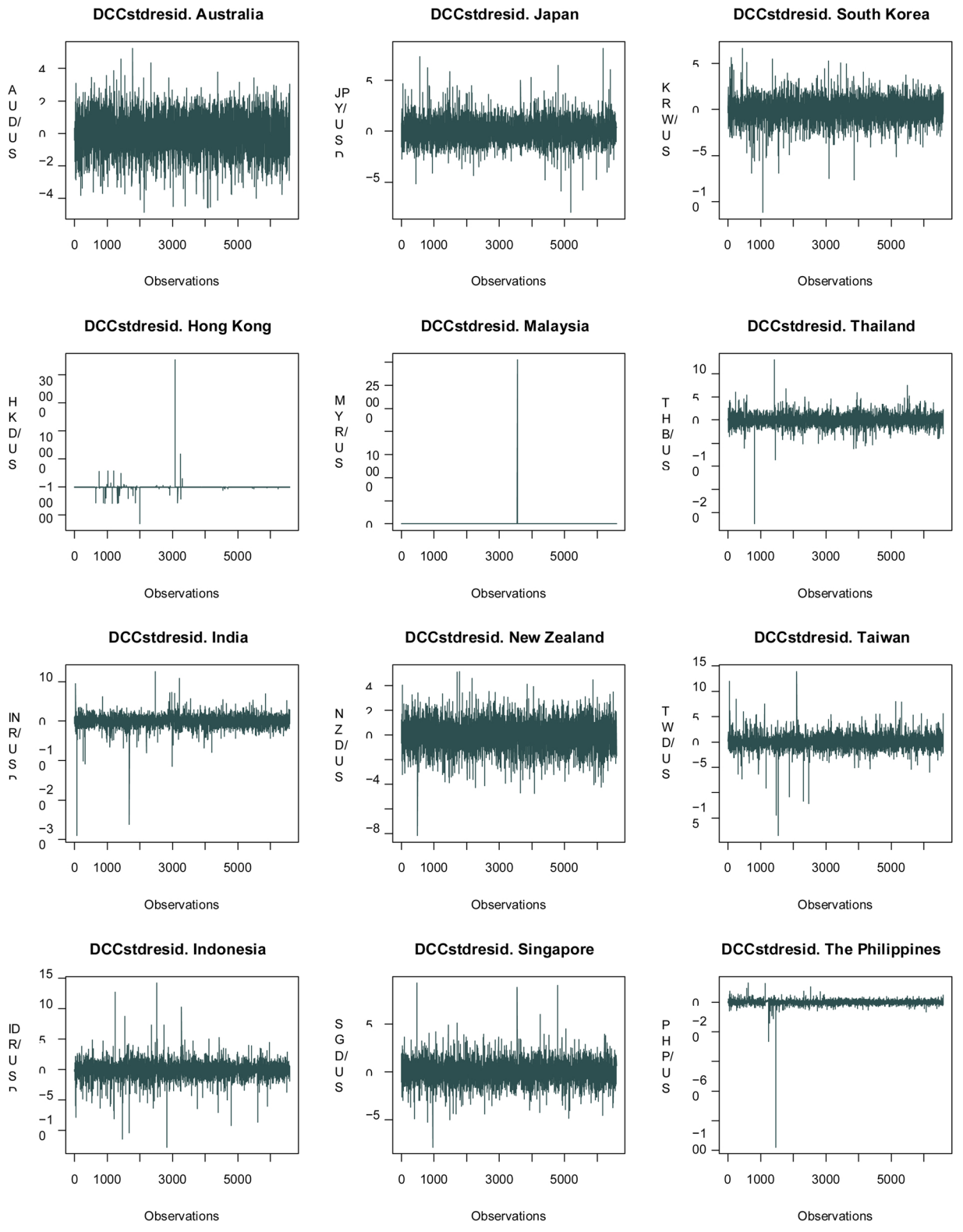


Fig. A2. DCC standard residuals.

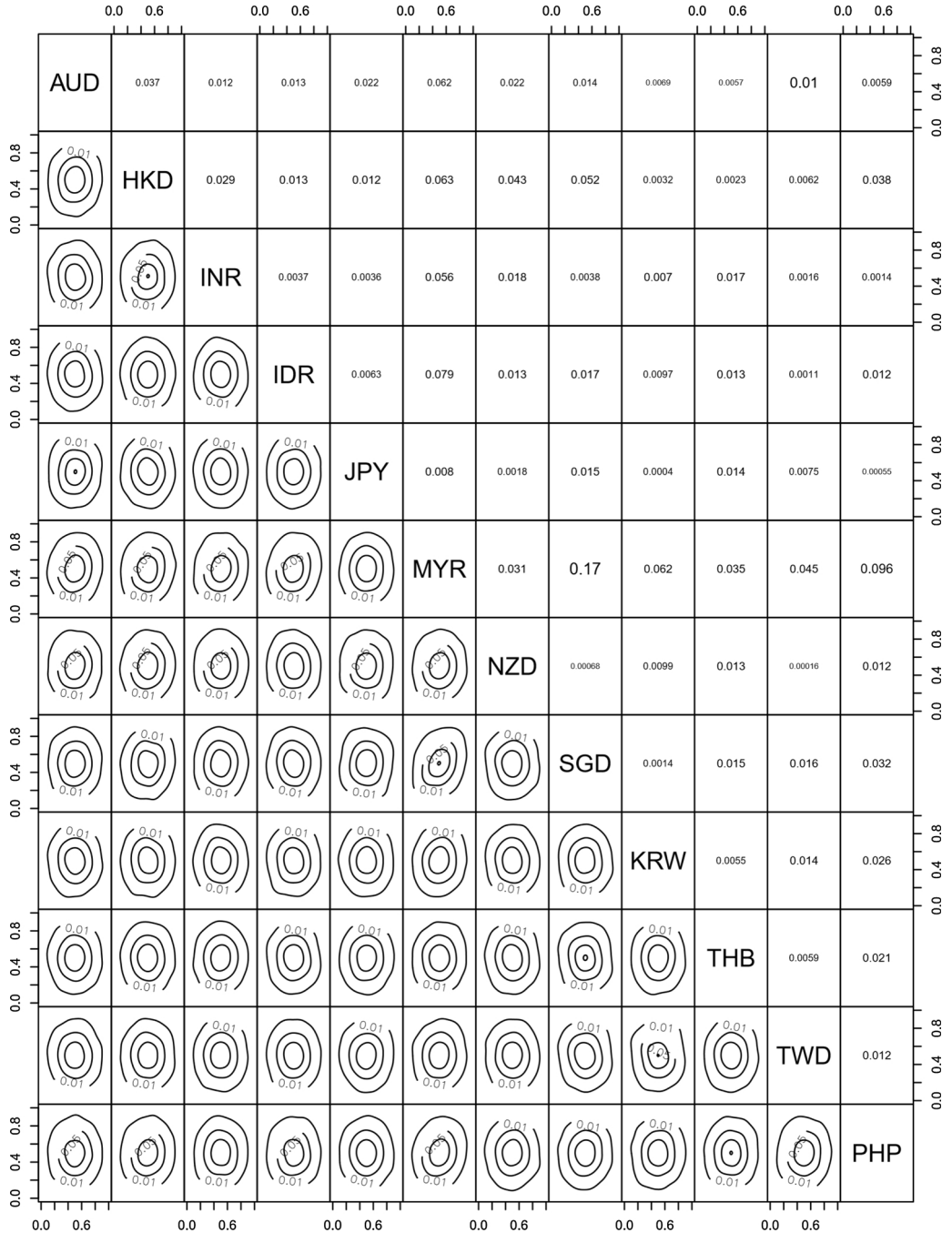


Fig. A3. Kendall's Taus and contours.

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Table A1
R-vine.

Tree†	Edge‡	No.	Family	Parameter	Parameter2	Tau
1	1,5	2	t	0.03	8.62	0.02
	6,10	9	BB7	1.06	0.02	0.04
	6,12	2	t	0.15	20.45	0.09
	2,7	20	SBB8	1.26	0.71	0.04
	6,1	2	t	0.09	30.00	0.06
	6,3	14	SG	1.06	0.00	0.06
	6,2	2	t	0.10	27.68	0.07
	6,4	17	SBB1	0.05	1.06	0.08
	6,9	2	t	0.10	16.16	0.06
	6,8	2	t	0.27	9.52	0.17
	11,6	19	SBB7	1.04	0.07	0.05
2	6,5|1	2	t	0.01	18.97	0.01
	8,10|6	2	t	−0.04	13.13	−0.02
	2,12|6	20	SBB8	1.15	0.76	0.03
	6,7|2	20	SBB8	1.06	0.96	0.03
	2,1|6	20	SBB8	1.12	0.87	0.03
	2,3|6	13	SC	0.04	0.00	0.02
	8,2|6	9	BB7	1.03	0.05	0.04
	8,4|6	2	t	−0.06	19.23	−0.04
	8,9|6	2	t	−0.03	17.47	−0.02
	11,8|6	2	t	−0.04	13.97	−0.02
3	2,5|6,1	5	F	−0.12	0.00	−0.01
	4,10|8,6	2	t	−0.03	15.45	−0.02
	3,12|2,6	2	t	−0.01	23.75	0.00
	1,7|6,2	2	t	0.02	14.65	0.01
	3,1|2,6	5	F	−0.15	0.00	−0.02
	8,3|2,6	2	t	−0.02	24.21	−0.01
	4,2|8,6	16	SJ	1.01	0.00	0.01
	9,4|8,6	2	t	−0.02	15.42	−0.02
	11,9|8,6	2	t	−0.02	9.38	−0.02
4	3,5|2,6,1	2	t	−0.00	29.59	−0.00
	2,10|4,8,6	6	J	1.01	0.00	0.01
	8,12|3,2,6	2	t	0.00	29.00	0.00
	3,7|1,6,2	3	C	0.02	0.00	0.01
	8,1|3,2,6	2	t	−0.01	13.06	−0.01
	4,3|8,2,6	33	C270	−0.02	0.00	−0.01
	9,2|4,8,6	2	t	0.00	26.75	0.00
	11,4|9,8,6	2	t	−0.01	28.76	−0.01
5	8,5|3,2,6,1	2	t	0.01	9.22	0.01
	3,10|2,4,8,6	14	SG	1.01	0.00	0.01
	1,12|8,3,2,6	0	I	-	-	0.00
	8,7|3,1,6,2	2	t	−0.02	22.70	−0.01
	4,1|8,3,2,6	5	F	−0.18	0.00	−0.02
	9,3|4,8,2,6	134	Tawn270	−1.07	0.06	−0.01
	11,2|9,4,8,6	4	G	1.01	0.00	0.01
6	4,5|8,3,2,6,1	2	t	0.01	30.00	0.01
	1,10|3,2,4,8,6	24	G90	−1.01	0.00	−0.01
	7,12|1,8,3,2,6	0	I	-	-	0.00
	4,7|8,3,1,6,2	2	t	−0.03	29.16	−0.02
	9,1|4,8,3,2,6	2	t	−0.02	27.56	−0.01
	11,3|9,4,,8,2,6	33	C270	−0.02	0.00	−0.01
7	10,5|4,8,3,2,6,1	5	F	0.12	0.00	0.01
	7,10|1,3,2,4,8,6	5	F	−0.15	0.00	−0.02
	4,12|7,1,8,3,2,6	2	t	0.00	20.73	0.00
	9,7|4,8,3,1,6,2	2	t	−0.01	30.00	−0.01
	11,1|9,4,8,3,2,6	224	Tawn2	90	−1.05	0.10-0.01
8	7,5|10,4,8,3,2,6,1	2	t	0.00	13.12	0.00
	12,10|7,1,3,2,4,8,6	2	t	0.02	22.28	0.02
	9,12|4,7,1,8,3,2,6	16	SJ	1.03	0.00	0.02
	11,7|9,4,8,3,1,6,2	0	I	-	-	0.00
9	12,5|7,10,4,8,3,2,6,1	0	I	-	-	0.00
	9,10|12,7,1,3,2,4,8,6	2	t	−0.01	30.00	−0.01
	11,12|9,4,7,1,8,3,2,6	2	t	0.01	28.18	0.01
10	9,5|12,7,10,4,8,3,2,6,1	0	I	-	-	0.00
	11,10|9,12,7,1,3,2,4,8,6	23	C90	−0.03	0.00	−0.01
11	11,5|9,12,7,10,4,8,3,2,6,1	13	SC	0.03	0.00	0.02

LogLik = 1081.76, AIC= -1955.53, BIC= -1248.98 ‡
1: Australia, 2: Hong Kong, 3: India, 4: Indonesia, 5: Japan, 6: Malaysia, 7: New Zealand, 8: Singapore, 9: South Korea, 10: Thailand, 11: Taiwan, 12: The Philippines.

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